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Planck hypothesis:

Planck assumed that the wall of black body ~~is~~ that is atoms and molecules on the wall ~~are~~ are "resonators" (Oscillators = Einstein).

"Energy of any such resonator is an integral multiple of the quantity $h\nu$; that is $E = nh\nu$, — (I)
where n is the integer"

"Energy is not emitted or absorbed continuously, but in discrete amounts (quanta), each of magnitude $\epsilon = h\nu$ "

According to Boltzmann's distribution, the probability of a mode with energy ϵ at temp T is $e^{-\epsilon/kT}$

$\therefore$ average energy of a mode is

$$\bar{\epsilon} = \frac{\sum_{n=0}^{\infty} \epsilon_n e^{-\epsilon_n/kT}}{\sum_{n=0}^{\infty} e^{-\epsilon_n/kT}} \quad \text{--- (II)}$$

$$= \frac{\sum_{n=0}^{\infty} nh\nu e^{-\frac{nh\nu}{kT}}}{\sum_{n=0}^{\infty} e^{-\frac{nh\nu}{kT}}} \quad \text{--- (III)}$$

$$\text{let } e^{-h\nu/kT} = x$$

$$\therefore \text{denominator } \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots \infty$$

$$= \frac{1}{1-x}$$

(2)

Numerator

$$\sum_{n=0}^{\infty} n x^n = x \frac{d}{dx} \sum_{n=0}^{\infty} x^n$$

$$= x \cdot \frac{d}{dx} \frac{1}{1-x}$$

$$= x \frac{1}{(1-x)^2}$$

$$(iii); \quad \bar{\epsilon} = \langle \epsilon \rangle = \frac{h\nu \cdot \frac{x}{(1-x)^2}}{\frac{1}{1-x}}$$

$$= h\nu \frac{x}{1-x}$$

$$= \frac{h\nu}{x^{-1} - 1}$$

dividing by $x^{\pm}$

$$= \frac{h\nu}{e^{h\nu/kT} - 1}$$

$$\boxed{\bar{\epsilon} = \langle \epsilon \rangle = \frac{h\nu}{e^{\frac{h\nu}{kT}} - 1}} \quad (iv)$$

Spectral Energy density $u(\nu, T) d\nu$
 [Energy per unit volume, per unit frequency at ν & T]
 $u(\nu, T) d\nu = \text{number of modes} \times \text{average energy per mode}$

$$u(\nu, T) d\nu = g(\nu) \langle E \rangle d\nu$$

$$= \frac{8\pi\nu^2}{c^3} \times \frac{h\nu}{e^{\frac{h\nu}{kT}} - 1} d\nu$$

$$\boxed{u(\nu, T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{\frac{h\nu}{kT}} - 1}} \quad \text{--- (V)}$$

in terms of wavelength; $\nu = \frac{c}{\lambda}$

$$d\nu = \left| -\frac{c}{\lambda^2} d\lambda \right|$$

$$\cancel{u(\lambda, T) d\lambda} \quad u(\lambda, T) d\lambda = \frac{8\pi h c}{\lambda^3 \cancel{\lambda^0}} \times \frac{c}{\lambda^2} d\lambda \cdot \frac{1}{e^{\frac{hc}{\lambda kT}} - 1}$$

$$\boxed{u(\lambda, T) d\lambda = \frac{8\pi h c}{\lambda^5} \cdot \frac{d\lambda}{e^{\frac{hc}{\lambda kT}} - 1}} \quad \text{--- (VI)}$$

Case 1 if $\lambda \rightarrow 0$
 $e^{\frac{hc}{\lambda kT}} \rightarrow \infty$
 $u(\lambda, T) \rightarrow 0$

i.e. there is no "ultraviolet catastrophe"

now for $e^{\frac{hc}{\lambda kT}} = 1 + \frac{hc}{\lambda kT} + \frac{1}{2} \left(\frac{hc}{\lambda kT} \right)^2 + \dots$

Case 2 for λ very large

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$$e^{\frac{hc}{\lambda kT}} = 1 + \frac{hc}{\lambda kT} + \frac{1}{2} \frac{h^2 c^2}{\lambda^2 k^2 T^2}$$

~ 0 for λ very very large

$$\approx 1 + \frac{hc}{\lambda kT}$$

$$\begin{aligned} \therefore u_{\lambda}(\lambda, T) d\lambda &= \frac{8\pi hc}{\lambda^5} \cdot \frac{d\lambda}{e^{\frac{hc}{\lambda kT}} - 1} \\ &= \frac{8\pi hc}{\lambda^5} \cdot \frac{d\lambda}{1 + \frac{hc}{\lambda kT} - 1} \\ &\approx \frac{8\pi hc}{\lambda^5} \cdot \frac{d\lambda}{\frac{hc}{\lambda kT}} \end{aligned}$$

$$\boxed{u_{\lambda}(\lambda, T) d\lambda \approx \frac{8\pi kT}{\lambda^4} d\lambda} \quad \text{--- (VIII)}$$

Rayleigh-Jeans formula

Case 3

Thus for very large λ (wavelength) plank radiation formula (spectral energy density $u_{\lambda}(\lambda, T) d\lambda$) reduces to R-J law.

Case 3: for $\lambda \ll \frac{hc}{kT}$ λ is very small

i.e. $\lambda \ll \frac{hc}{kT}$

$$e^{\frac{hc}{\lambda kT}} \gg 1$$

(5)

Then $u_{\lambda}(\lambda, T) d\lambda = \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{e^{\frac{hc}{\lambda kT}} - 1}$

$$= \frac{8\pi hc}{\lambda^5} \frac{d\lambda}{e^{\frac{hc}{\lambda kT}}}$$

$$= \frac{8\pi hc}{\lambda^5} e^{-\frac{hc}{\lambda kT}} d\lambda$$

$$\boxed{u_{\lambda}(\lambda, T) d\lambda = \frac{A}{\lambda^5} e^{-B/\lambda T} d\lambda}$$

this form of

Wein's law
thus for smaller wavelength plank's law
reduces to Wein's law.

Total Energy density of a Black Body

$$u = \int_0^{\infty} u_{\lambda} d\lambda$$

$$= \int_0^{\infty} \frac{8\pi hc}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda kT}} - 1} d\lambda$$

Let $\frac{hc}{\lambda kT} = x$

$$\lambda = \frac{hc}{x kT} \quad \therefore d\lambda = \frac{-hc}{kT x^2} dx$$

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$$\begin{aligned}
 \therefore u &= \int_0^\infty \frac{8\pi hc}{(hc)^5} \cdot (\pi kT)^5 \cdot \frac{1}{e^\pi - 1} \cdot \frac{-hc}{kT\pi^2} d\pi \\
 &= \int_0^\infty \frac{8\pi kT^4}{h^3 c^3} \cdot \frac{\pi^3}{e^\pi - 1} d\pi \\
 &= \frac{8\pi kT^4}{h^3 c^3} \int_0^\infty \frac{\pi^3}{e^\pi - 1} d\pi \\
 &= \frac{8\pi kT^4}{h^3 c^3} \times \frac{\pi^4}{15}
 \end{aligned}$$

$\lambda \rightarrow 0; \pi \rightarrow \infty$
 $\lambda \rightarrow \infty; \pi \rightarrow 0$
 so limit changes
 to remove
 $-hc \rightarrow hc$
 limit again
 changes

$$u = \left[\frac{8\pi^5 k^4}{15 h^3 c^3} \right] T^4$$

~~in terms of frequency~~

$$u(T) = a T^4$$

where

$$a = \left(\frac{8\pi^5 k^4}{15 h^3 c^3} \right)$$

$$= 7.5657 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4}$$

check.

$$= 7.5657 \times 10^{-16} \text{ J m}^{-3} \text{ K}^{-4}$$

$$\begin{aligned}
 1 \text{ W} &= 1 \text{ per second} \\
 &= 1 \text{ J s}^{-1}
 \end{aligned}$$

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Deduction of Stefan's-Boltzmann's law

The power radiated per unit area of a Black body is given

$$J = \sigma T^4$$

where

σ is Stefan's constant

Emissive power

Emissive power

$$J = \frac{c}{4} U$$

$$J = \frac{c}{4} U(T) \quad ; \quad U_\lambda \text{ or } U_\nu$$

$$= \frac{c}{4} \left(\frac{8}{15} \cdot \frac{\pi^5 k^4}{h^3 c^3} \right) T^4 \quad \text{both are same here}$$

$$= \frac{2}{15} \frac{\pi^5 k^4}{h^3 c^2} T^4$$

$$\boxed{J = \sigma T^4}$$

$$\text{Where } \sigma = \frac{2}{15} \frac{\pi^5 k^4}{h^3 c^2}$$

$$= \text{W m}^{-2} \text{K}^{-4}$$

$$= (\text{J s}^{-1}) \cdot \text{m}^{-2} \text{K}^{-4}$$

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